

Multi-Dataset Diagnostic Utility of Clinical Visual Concepts in AI Systems for Dermatology

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Abstract

The Challenge

The clinical integration of AI in digital dermatology relies heavily on human trust, where interpretable visual concepts can act as intermediate representations. However, progress is hindered by scattered, heterogeneous annotations across existing datasets.

Our Approach: SkinLex & Concept Utility

We introduce SkinLex, a harmonized dataset of 48 clinical attributes across 20,411 samples grounded in human expertise. We evaluate concept utility for 9-partition disease classification.

Key Outcomes

- Restricting inputs to specific concept groups (e.g., shapes or colors alone) significantly degrades diagnostic accuracy.
- Bootstrapped backward elimination reveals redundant concepts for 9-partition diagnosis, allowing omission of 8-18 concepts without severe performance loss.

Introduction

Visual clinical concepts (Fig. 1) enhance trust in AI decision-support systems, but analyzing their utility is limited by non-standardized public annotations.



(a) **Malignant Melanoma**
Papule, Erythema



(b) **Inflammatory**
Papule, Plaque, Xerosis,
Scale, Purple



(c) **Inflammatory**
Plaque, Erythema

Figure 1. Example lesions [1] with concepts [2] and 9-partition categories. Image courtesy of AtlasDermatologico.com.br / Dr. Samuel Freire da Silva

Methods

Dataset Harmonization

SkinLex (Fig. 2) aggregates samples across four datasets, combining direct concept labels (SkinCon [2], DermaCon-IN [3]) with concepts extracted from captions via Qwen3.6 (MM-Skin [4], PASSION [5]). Granular diseases were mapped to a 9-partition scheme [1] using Qwen.

Experimental Setup

We evaluate 48 concepts for 9-partition diagnosis using LightGBM and Logistic Regression across 100 bootstraps, examining standalone concept groups, leave-one-out sensitivity, and backward elimination.

References & Notes

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- [2] R. Daneshjou, M. Yuksekgonul, Z. R. Cai, et al., "SkinCon: A skin disease dataset densely annotated by domain experts for fine-grained debugging and analysis," in *NeurIPS*, vol. 35, 2022.
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- [4] W. Zeng, Y. Sun, C. Ma, et al., "MM-Skin: Enhancing Dermatology Vision-Language Model with an Image-Text Dataset Derived from Textbooks," in *Proc. ACM Multimedia*, 2025.
- [5] P. Gottfrois, F. Gröger, F. H. Andriambololoniaina, et al., "PASSION for Dermatology: Bridging the Diversity Gap with Pigmented Skin Images from Sub-Saharan Africa," in *Proc. MICCAI*, 2024.

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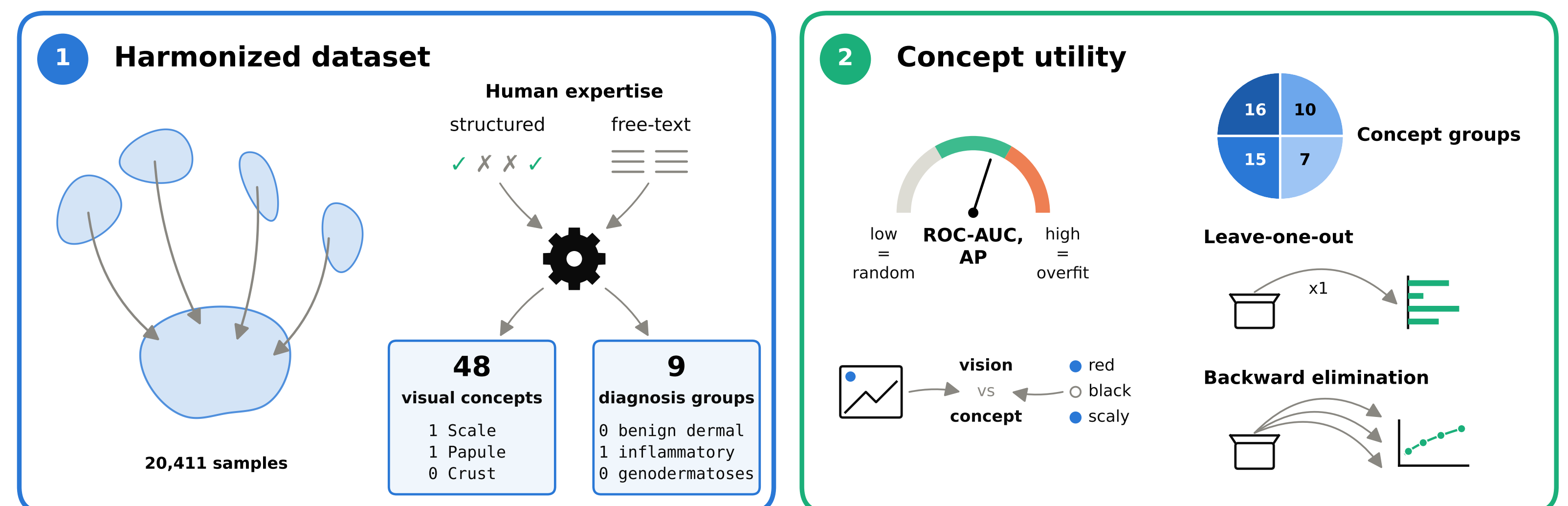


Figure 2. SkinLex harmonization pipeline (left) and evaluation framework (right).

Experiments & Results

LightGBM achieves the highest performance among tabular classifiers, capturing 90.9% of the ROC-AUC ceiling (0.8555) set by vision backbones. Logistic Regression serves as a linear baseline.

Concept Groups & Sensitivity Analysis

Restricting inputs to specific concept groups degrades performance (Table 1). Leave-one-out sensitivity highlights that *Brown (Hyperpigmentation)* causes the largest degradation when removed, followed by structural attributes (*Plaque, Papule, Nodule, Scale*).

Table 1. Model performance across full and restricted concept groups (median 95% CI).

Feature Set	Features	LightGBM		Logistic Regression	
		ROC-AUC	AP	ROC-AUC	AP
Full Model	48	0.829 [0.845, 0.812]	0.309 [0.330, 0.288]	0.801 [0.819, 0.786]	0.274 [0.299, 0.255]
Shapes	16	0.711 [0.727, 0.695]	0.174 [0.183, 0.163]	0.672 [0.689, 0.658]	0.170 [0.181, 0.158]
Colors	10	0.690 [0.713, 0.671]	0.173 [0.185, 0.164]	0.683 [0.707, 0.665]	0.177 [0.191, 0.163]
Textures	15	0.661 [0.675, 0.644]	0.153 [0.163, 0.143]	0.659 [0.673, 0.643]	0.155 [0.163, 0.148]
Others	7	0.530 [0.538, 0.523]	0.124 [0.129, 0.119]	0.532 [0.541, 0.526]	0.124 [0.129, 0.119]

Concept Redundancy via Backward Elimination

Bootstrapped backward elimination shows that performance remains stable even as multiple concepts are removed (Fig. 3). Paired Wilcoxon signed-rank tests ($p < 0.05$) confirm that up to 8 concepts can be omitted without a significant drop in AP, and up to 18 for ROC-AUC.

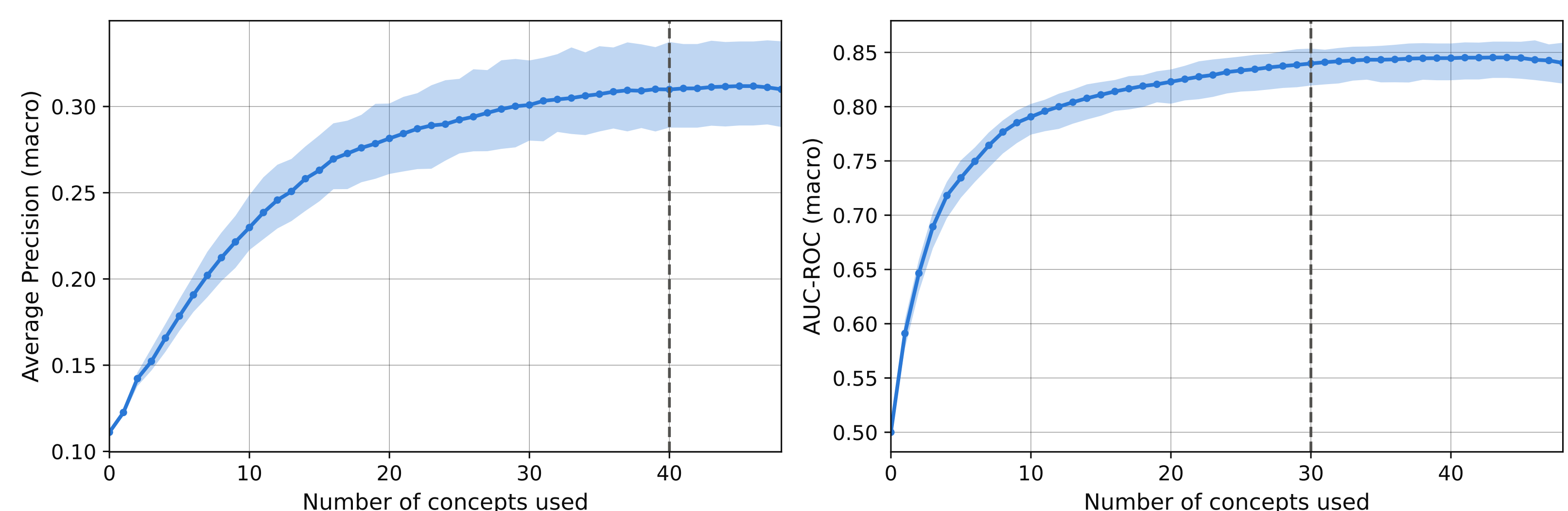


Figure 3. LightGBM performance (macro AP and ROC-AUC) relative to the number of retained concepts across 100 bootstrap samples.

Conclusion

Visual clinical concepts provide interpretable intermediate representations for AI decision support. On SkinLex, models omit 8 to 18 concepts without losing accuracy, showing feature redundancy in 9-partition diagnosis. While individual concepts may not be universally redundant, refining concept sets could lower annotation and computational burden. This work provides a foundation for optimizing dermatological taxonomies.